# Propositional Logic Homework a, b, c, d: Each 10 points

## (a) Prove use words:

$$(a \vee b) \wedge (\neg a \vee c) \rightarrow (b \vee c)$$

## (b)

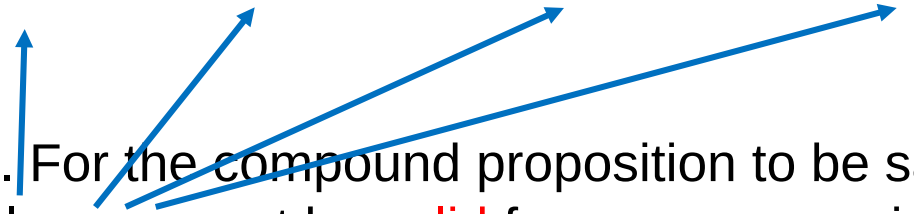


2. *Propositional Resolution.* Use propositional resolution to show that the following sets of clauses are unsatisfiable.
   (a) $\{p,q\}, \{\neg p,r\}, \{\neg p,\neg r\}, \{p, \neg q\}$ ← Proof is given in following slide
   (b) $\{p,q, \neg r,s\}, \{\neg p,r,s\}, \{\neg q, \neg r\}, \{p, \neg s\}, \{\neg p, \neg r\}, \{r\}$ ← You prove this

Note {p,q} = p V q is a clause. We use "~" for negation operator, "&" for AND operator.
Thus (a) means show (p V q) & (~p V r) & (~p V ~r) & (p V ~q) is unsatisfiable.

Use resolution to prove that

(p V q) & (~p V r) & (~p V ~r) & (p V ~q) is unsatisfiable.



Proof. For the compound proposition to be satisfiable, each of the four clauses must be valid for a proper assignment of p,q,r. Thus by resolution, we have

By distribution, A&B&C&D = D&A&B&C
Thus the above compound proposition becomes

（p V ~q) & (p V q) & (~p V r) & (~p V ~r)
= [ (p V ~q) & (p V q) ] & [ (~p V r) & (~p V ~r) ]
= [ p V p ] & [ ~p V ~p ]  do resolution on both
= [p] & [~p]  why?
= false  why?

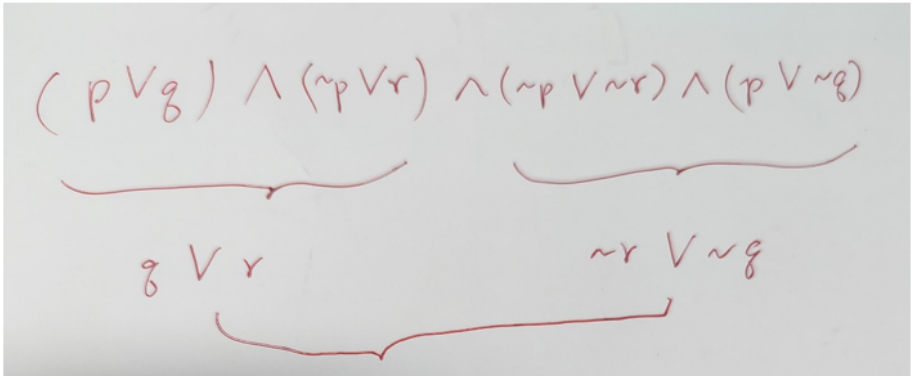
Thus, there is no assignment that makes the proposition true. It is unsatisfiable.

**This is another attempt to prove. It did not provide useful things.**

Use resolution to prove that

(p V q) & (~p V r) & (~p V ~r) & (p V ~q) is unsatisfiable.

Proof. For the compound proposition to be satisfiable, each of the four clauses must be valid for a proper assignment of p,q,r. Thus by resolution, we have




r V ~r = true

We start with the assumption that the whole compound proposition is true. We end up with it is true. We did not prove/get anything new. The required task is starting with the assumption and then find a contradiction – which show the assumption is invalid.

# Propositional Logic Homework: (C)

**58.** How many of the disjunctions $p \vee \neg q$, $\neg p \vee q$, $q \vee r$, $q \vee \neg r$, and $\neg q \vee \neg r$ can be made simultaneously true by an assignment of truth values to $p$, $q$, and $r$?

# Propositional Logic Homework (D)

In standard mathematics, can the following three equations hold (being true) simultaneously?

$$x = y$$
$$y = z$$
$$z = -x$$

Why?

In propositional logic, consider the following three compound propositions

$$A \Rightarrow B$$
$$B \Rightarrow C$$
$$C \Rightarrow \neg A$$

(a) They can not be true simultaneously in all cases?
(b) They can be true simultaneously in some cases?
(c) They can be true simultaneously in all cases?
Provide your answer. Explain or prove it.
(Hint: use the truth table to guide your thinking.)